6. Prove the following Boolean expressions using the laws and theorems of Boolean algebra.

i) $(A+B)\,(A+\overline{B}) = A$

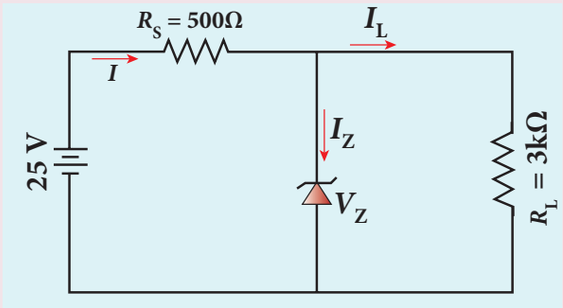
ii) $A(\overline{A}+B) = AB$

iii) $(A+B)\,(A+C) = A+BC$

7. Verify the given Boolean equation $A + \overline{A}B = A + B$ using truth table.

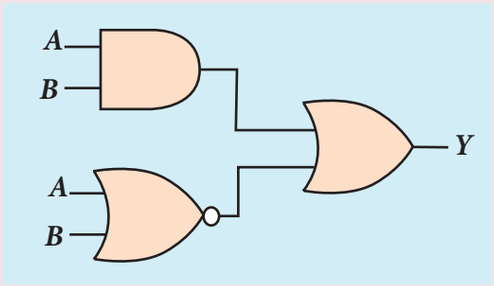
8. In the given figure of a voltage regulator, a Zener diode of breakdown voltage 15V is employed. Determine the current through the load resistance, the total current and the current through the diode. Use diode approximation.

[Ans: 5mA; 20 mA; 15 mA]



9. Write down Boolean equation for the output Y of the given circuit and give its truth table.

[Ans: $Y = (AB)+(\overline{A+B})$]



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